Student Name: _____ ID: _____

CSE2046/CSE2246 Analysis of Algorithms

Spring 2022 Quiz – Group A

Duration: 40 minutes

Q-1. Multiple Choice (75 pts): For each false answer: -2 pts

1) (8 pts) Which class is correct for $f(n) = 8^{n/2} + nlogn$?

a) $\Theta(2^n)$ b) $O(2^n)$ c) $\Theta(nlogn)$ d) $O\left((2\sqrt{2})^n\right)$ e) $\Omega(8^n)$

2) (8 pts) Which of the following Is false?

a) $\log(n)$ has lower asymptotical order than $(\log n)^2$.
b) $\log_3 n$ and $\ln n$ are of the same asymptotical order.
c) $(n+1)!$ has the same asymptotic growth as $n!$.
d) Asymptotic order of a polynomial with degree n is always higher than asymptotical order of a polynomial with degree less than n.
e) $(1{,}8)^n$ has higher asymptotic order than n^{100}.

3) (8 pts) Which of the following is false?

a) $\log_2(n^2+1) \in O(n)$
b) $2 \cdot \log_3 \sqrt[3]{n} \in \Theta((\log n)$
c) If $f(n) \in O(g(n))$ then $g(n) \in \Omega(f(n))$
d) For any function $f(n)$, it is true that $f(2n) \in O(f(n))$ e.g. $(2n)! \notin O(n!)$
e) For any function $f(n)$, it is true that $f(n) \in \Theta(2f(n))$

4) (8 pts) Which of the following functions has the highest asymptotic growth?

a) $a(n) = (n/2)!$
b) $b(n) = 8^n + n^8$
c) $c(n) = 10^{n/2} + n^5$
d) $f(n) = n\log(n^n)$
e) $g(n) = n2^{3n}$

5) (8 pts) Give a recurrence relation for the number of multiplications $M(n)$ performed in the following function?

```
function g(n)

if n <= 2:
      return 5
else:
      return g(n-1)*g(n-2)
```

a) $M(n) = M(n-1) * M(n-2); M(1) = M(2) = 5$
b) $M(n) = M(n-1) + M(n-2); M(1) = M(2) = 5$
c) $M(n) = M(n-1) + M(n-2) + 1; M(1) = M(2) = 5$
d) $M(n) = M(n-1) + M(n-2); M(1) = M(2) = 0$
e) $M(n) = M(n-1) + M(n-2) + 1; M(1) = M(2) = 0$

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Consider the following algorithm. Answer questions 6 and 7.

```
void func1(int n) {
      int i = 1;
      int x = 0;
      int count = 0;

      while (i <= n){
            x = x + 2;
            i = i*3;
      }
      for(int j = 1; j <= x; j++)
            for(int k = 1; k <= j; k++)
                  for(int t = 1; t <= n; t++)
                        count = count + 1;
}
```

6) (8 pts) What is the final value of x?

a) $6n$ b) $2\lceil \log_3 n \rceil$ c) $3\lceil \log_2 n \rceil$ d) 2n/3 e) $2n^3$

7) (8 pts) What is the time complexity of func1 given above?

a) $\Theta(n^3)$ $b)\ \Theta(nlog^2n)$ $c)\ \Theta(n^2logn)$ $d)\ \Theta(log^2n)$ $e)\ \Theta(log^3n)$

8) (8 pts) Which of the following is a correct recurrence relation for the number of lines the function prints?

```
function f(n)
if n > 1:
      for i=1 to n {
            f(n-1);
            print_line("Take it easy") ;
      }
```

a) $P(n) = nP(n-1); P(1) = 1;$
b) $P(n) = nP(n-1) + 1; P(1) = 0;$
c) $P(n) = P(n-1) + n; P(1) = 0;$
d) $P(n) = nP(n-1) + n; P(1) = 0;$
e) $P(n) = nP(n-1) + n; P(1) = 1;$

9) (11 pts) What would be the solution of the following recurrence relation?

$$T(n) = 2T(n-1) + 2; \quad T(1) = 1$$

a) $T(n) = 2^{n-1} - 1$
b) $T(n) = 2^n - 1$
c) $T(n) = 3 \cdot 2^{n-1} - 1$
d) $T(n) = 2^{n+1} - 1$
e) $T(n) = 3 \cdot 2^n - 1$

$T(n) = 2(2T(n-2)+2)+2 = 2^2T(n-1)+2^2+2$
$= 2^2(2T(n-3)+2)+2^2+2 = 2^3T(n-1)+2^2+2^2+2$
$= 2^{n-1}T(1)+2^{n-1}+2^{n-2}+..+2^2+2$
$= 2^{n-1}+2^n-1-1 = 3.2^{n-1}-2$

Q-2. (30 pts) A, B and C are three integer arrays of size m, n and t respectively. You are asked to print all the integers that exist in A, but neither in B nor in C (i.e. $(A-B)-C$). Write a pseudocode for a brute-force algorithm and give its time complexity (in worst and best case). Please give your answer in a separate blank sheet.

```
for i = 1 to m {
    inB ← false ; inC ← false
    for j=1 to n
        if A[i] == B[j] {
            inB ← true
            continue;
        }
    if (inB == false)
        for k=1 to t
            if A[i] == C[k] {
                inC ← true
                continue;
            }
    if (inB == false) && (inC == false)
        print(A[i]);
}
```

Best Case Complexity = $\Theta(m)$

Worst Case " = $\Theta(mn+mt)$

In the best case, all the elements of A are the same and equal to the first element of B.

In the worst case, all the elements of A, B & C are different.